

Stability Analysis for Decentralized Control of Multi-Evaporator Vapor-Compression Cycle Systems

Neera Jain, Shreyas Sundaram, and Andrew G. Alleyne

Abstract—We consider the problem of stabilizing multi-evaporator vapor-compression cycle (ME-VCC) systems using decentralized controllers. ME-VCC systems, sometimes termed variable-refrigerant-flow systems, are prevalent in large buildings that maintain independent cooled spaces with a single heat rejection unit. We exploit the time-scale separation characteristic of ME-VCC systems and analyze the faster mass flow dynamics and their stability characteristics independently of the slower thermal dynamics in the system. An electrical circuit analogy is used to obtain a linearized state-space representation of the mass flow dynamics for two common architectures of ME-VCC systems. Using concepts from decentralized control theory, we provide conditions under which local static feedback controllers stabilize the overall closed-loop system with robustness to uncertainties in the coupling between subsystems. Our analysis characterizes the beneficial impact that discharge pressure regulating (DPR) valves have on the decentralized controller gains.

I. INTRODUCTION

In an effort to reduce energy use in buildings, there is a growing interest in the efficiency of large-scale heating and cooling systems. In particular, multi-evaporator vapor-compression cycle (ME-VCC) systems, also termed variable-refrigerant-flow (VRF) systems, are gaining significant market attention in the U.S. because of the benefits they offer with respect to energy efficiency and flexibility to building architects [1] [2].

ME-VCC systems consist of a single condenser (heat rejection unit) which is then connected to several evaporators (heat absorbing units). For every evaporator, there is an accompanying electronic expansion valve (EEV) which is actuated to vary the amount of refrigerant flow through that evaporator, thereby affecting the amount of cooling delivered to a particular space or zone.

It is common to have more than 30 evaporators interacting with the single condenser and compressor in a ME-VCC system [1]. Each zone in a building typically has a dedicated evaporator and EEV, so a decentralized control architecture is more practical as the number of evaporators increases. Although the dynamics of ME-VCC systems are highly nonlinear, linearized models are typically used for the

purpose of control design. Fortunately, decentralized controllers have the potential to provide robustness to the parametric uncertainties present in any linearized model.

The fact that all n evaporators are interacting with a single compressor and condenser introduces significant coupling in the system. Each evaporator unit is tasked with meeting its own performance demand but is impacted by the demands on the other evaporators in the overall network. The objective of this paper is to show conditions under which stabilizing decentralized controllers can be designed for arbitrarily large ME-VCC systems.

The first challenge in controlling ME-VCC systems is the development of an appropriate model of the system dynamics. System identification techniques are typically employed to obtain black-box models mapping a prescribed set of inputs to a prescribed set of outputs. The inputs and outputs are chosen based on conventional control of the system and/or physical intuition [3] [4]. However, this precludes the possibility of finding alternative input-output pairings which might offer advantages from a control systems perspective [5]. Additional disadvantages of a data-driven model are that 1) it cannot be used during the system design phase since the use of model identification techniques requires that the system already exists, and 2) it is not scalable (with respect to the number of evaporators in the system) since it will be specific to the system configuration at the time the data is gathered.

Various control schemes have been proposed for dual- and triple-evaporator VCC systems. In [4] a cascaded control structure is proposed for a dual-evaporator VCC system which takes advantage of the time-scale separation between the refrigerant thermal dynamics and the zone air thermal dynamics. A stability analysis motivates the choice of controller gains for individual PI controllers. The paper [6] introduces a fuzzy algorithm used with PD controllers to regulate zone air temperatures for a triple-evaporator VCC system. In [7] a decentralized controller with a supervisory layer is used to achieve a desired cooling capacity in each zone while also minimizing energy use in a dual-evaporator VCC system.

What is lacking in the literature is a scalable analysis of arbitrarily large ME-VCC systems. The problem of controlling ME-VCC systems with $n = 2$ is already quite challenging and, in practice, these systems typically exist with $n \gg 2$. In this paper we use concepts from decentralized control theory to analyze and design local static feedback controllers for each evaporator so that the overall closed-loop system is stable and robust to uncertainties in the coupling between subsystems. This analysis is enabled by a novel modeling approach in which

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we exploit the time-scale separation characteristic of ME-VCC systems [8] and analyze the faster mass flow dynamics and their stability characteristics independently of the slower thermal dynamics in the system. This sufficiently reduces the complexity of the problem so that a scalable analysis is possible. Using an electrical circuit analogy, we model only the mass flow dynamics and allow the physics to guide a choice of input-output pairings for control of these fast dynamics in the ME-VCC system.

We will study two common architectures for ME-VCC systems. The primary difference between the two architectures is that Architecture II is augmented with additional discharge pressure regulating (DPR) valves. DPR valves are typically included in ME-VCC systems, despite the additional cost of the hardware, to decrease the coupling between individual evaporators. However, for various reasons, DPR valves may not be an option or desirable for a particular ME-VCC system. Therefore, we are interested in understanding how the physical system architecture impacts stabilizability in both cases. In particular, our analysis will characterize the beneficial impact that DPR valves have on the decentralized controller gains.

This paper is organized as follows. In Section II, the dynamic model for each of two system architectures is derived. In Section III, we provide conditions under which stabilizing decentralized controllers can be designed for arbitrarily large ME-VCC systems. Finally, our conclusions are summarized in Section IV.

II. DYNAMIC MODELING

The two system architectures we will consider are shown in Fig. 1 and Fig. 2, respectively. Architecture II is identical to Architecture I with the addition of discharge pressure regulating (DPR) valves at the outlet of $n-1$ evaporators.¹

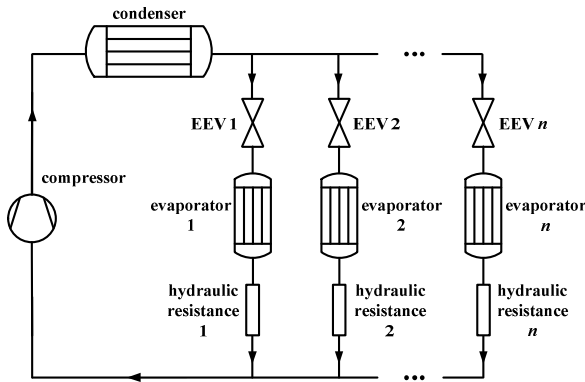


Fig. 1. Schematic of ME-VCC system Architecture I.

An electrical circuit analogy is used to model the mass flow dynamics of each ME-VCC system in terms of capacitors, variable resistors and standard resistors, and a single current source for an arbitrarily large number (n) of evaporators. An advantage of this gray-box modeling approach is that the functional relationships between the

¹ Typically, the pressure of the first evaporator, P_1 , is the pressure of the combined mass flow at the compressor inlet in Architecture II. Therefore, the hydraulic resistance located downstream of evaporator 1 in Fig. 2 is assumed to have a zero resistance.

physical parameters are conserved. A linearized model is derived; therefore, all variables in the subsequent derivation represent deviation variables with respect to some nominal operating condition.

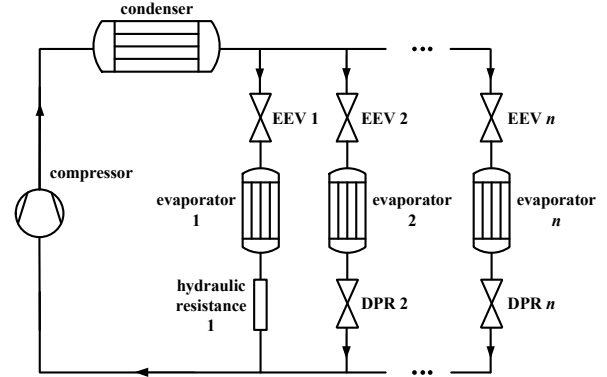


Fig. 2. Schematic of ME-VCC system Architecture II.

A. Architecture I

The electrical circuit analogy for Architecture I is shown in Fig. 3. The component-level analogues are described in Table I.

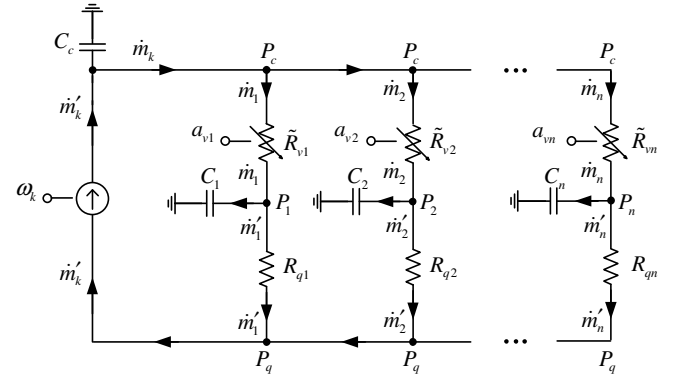


Fig. 3. Electrical circuit analogy of Architecture I. The symbol $\circ$ — indicates the presence of a control input.

TABLE I
ELECTRICAL CIRCUIT ANALOGUES FOR ME-VCC SYSTEMS

ME-VCC System	Electrical Circuit Analogue
compressor	current source
EEV / DPR valve	variable resistor
hydraulic resistance	resistor
condenser/evaporators	capacitor
refrigerant mass flow rate	electric current
pressure differential	voltage potential

There are $n+1$ differential equations which correspond to the $n+1$ heat exchangers in the system – one condenser and n evaporators – as shown in (1). The dynamic states of the ME-VCC system are the refrigerant pressures, P_c and P_i , $i=1, \dots, n$.

$$\begin{aligned}
 C_c \dot{P}_c &= (\dot{m}'_k - \dot{m}_k) \\
 C_1 \dot{P}_1 &= (\dot{m}_1 - \dot{m}'_1) \\
 C_2 \dot{P}_2 &= (\dot{m}_2 - \dot{m}'_2) \\
 &\vdots \\
 C_n \dot{P}_n &= (\dot{m}_n - \dot{m}'_n)
 \end{aligned} \tag{1}$$

In (1), $\dot{P}_i$ is the derivative of the state, not to be confused with $\dot{m}_i$ which does not represent a state derivative but instead the refrigerant mass flow rate, an algebraic quantity in this system.

The compressor is modeled as a current source which provides a refrigerant mass flow rate $\dot{m}'_k$ to the system. The mass flow rate, given by

$$\dot{m}'_k = -\beta_{k1}P_q + \beta_{k2}P_c + \beta_{k3}\omega_k, \quad (2)$$

is a function of the pressures at the inlet and outlet of the compressor as well as the compressor speed, ω_k (a control input to the system). The parameters β_{k1} , β_{k2} , and β_{k3} , are linearization coefficients.

Each electronic expansion (control) valve (EEV) is modeled as a variable resistor whose resistance $\tilde{R}_{vi}$ is a function of the valve aperture, a_{vi} (a control input to the system). Thus, from Fig. 3 we obtain

$$\begin{aligned} P_c - P_1 &= \tilde{R}_{v1}(a_{v1})\dot{m}_1 \\ P_c - P_2 &= \tilde{R}_{v2}(a_{v2})\dot{m}_2 \\ &\vdots \\ P_c - P_n &= \tilde{R}_{vn}(a_{vn})\dot{m}_n. \end{aligned} \quad (3)$$

For simplicity, the equations in (3) are linearized so that the total pressure drop across each valve is expressed as a linear combination of the mass flow rate through the valve and the valve aperture as shown in (4). Both R_{vi} and K_{vi} are positive constants arising from the linearization and are functions of $\tilde{R}_{vi}$; R_{vi} can be thought of as the nominal resistance associated with the valve.

$$\begin{aligned} P_c - P_1 &= R_{v1}\dot{m}_1 - K_{v1}a_{v1} \\ P_c - P_2 &= R_{v2}\dot{m}_2 - K_{v2}a_{v2} \\ &\vdots \\ P_c - P_n &= R_{vn}\dot{m}_n - K_{vn}a_{vn} \end{aligned} \quad (4)$$

Following Kirchhoff's current law, the n refrigerant streams exiting the evaporators will combine at each of the circuit junctions. As shown in Fig. 3, the combined refrigerant stream is at a common pressure, P_q , where

$$P_q \leq P_i \quad \forall i = 1, \dots, n.$$

We model the pressure drop downstream of each evaporator using a resistor with resistance R_{qi} , $i = 1, \dots, n$, as shown in (5). We assume that the refrigerant exiting each of the evaporators is a single-phase vapor.

$$\begin{aligned} P_1 - P_q &= R_{q1}\dot{m}'_1 \\ P_2 - P_q &= R_{q2}\dot{m}'_2 \\ &\vdots \\ P_n - P_q &= R_{qn}\dot{m}'_n \end{aligned} \quad (5)$$

Finally, we can write two mass flow conservation equations for the ME-VCC modeled in Fig. 3 which are given by

$$\begin{aligned} \dot{m}'_k &= \dot{m}'_1 + \dot{m}'_2 + \dots + \dot{m}'_{n-1} + \dot{m}'_n \\ \dot{m}_k &= \dot{m}_1 + \dot{m}_2 + \dots + \dot{m}_{n-1} + \dot{m}_n. \end{aligned} \quad (6)$$

The governing equations have been written so that all of the system capacitances, resistances, and linearized coefficients are always positive. The resulting state-space system representation $\mathbf{S}_1: \delta \dot{\mathbf{x}}_1 = \mathbf{A}_1 \delta \mathbf{x}_1 + \mathbf{B}_1 \delta \mathbf{u}_1$,

$\mathbf{A}_1, \mathbf{B}_1 \in \mathbb{R}^{(n+1) \times (n+1)}$ is shown in (21)² where

$$\begin{aligned} \delta \mathbf{x}_1 &= [P_c \quad P_1 \quad P_2 \quad \dots \quad P_n]^T \\ \delta \mathbf{u}_1 &= [\omega_k \quad a_{v1} \quad a_{v2} \quad \dots \quad a_{vn}]^T. \end{aligned} \quad (7)$$

Recall that the states and inputs are deviation variables with respect to the point about which the system is linearized. The variable M in (21) is defined as

$$\frac{1}{M} = \left(\sum_{i=1}^n \frac{1}{R_{qi}} \right) - \beta_{k1}.$$

Now we provide an important result regarding the controllability of the model which we have just derived.

Proposition 1: The linear state-space model for Architecture I given in (21) is not controllable.

Proof: It can be verified that $\text{rank}(\mathbf{A}_1) < n+1$, indicating the existence of an eigenvalue at zero. Moreover, one can verify that the vector

$$\mathbf{v} = \begin{bmatrix} 1 & \frac{C_1}{C_c} & \frac{C_2}{C_c} & \dots & \frac{C_n}{C_c} \end{bmatrix}$$

is in the left nullspace of the matrix $[\mathbf{A}_1 - \lambda \mathbf{I} \mid \mathbf{B}_1]$ for $\lambda = 0$. Therefore, the eigenvalue at zero is not controllable. ■

Remark 1: The physical interpretation for this result can be obtained upon closer examination of (1) and (6). By virtue of the mass flow conservation equations (6), the sum of the $n+1$ differential equations in (1) is $\dot{P}_c = -\sum_{i=1}^n C_i C_c^{-1} \dot{P}_i$. It follows that $P_c = \left(-\sum_{i=1}^n C_i C_c^{-1} P_i \right) + \gamma$ where γ is a constant of integration which depends on the initial state of the system, related to system refrigerant charge. Here we assume that the initial state of the system is such that $\gamma = 0$. ■

Since the condenser pressure can be expressed as a linear combination of the n evaporator pressures, we eliminate P_c as a state in the system model. Moreover, since $\text{rank}(\mathbf{B}_1) = n$ in (21), only n of the $n+1$ inputs independently affect the n states. An appealing choice is to eliminate the compressor speed, ω_k , from the input vector $\delta \mathbf{u}_1$ (7) because this results in a diagonal matrix. With an abuse of notation, we will denote the revised state-space representation as $\mathbf{S}_1: \delta \dot{\mathbf{x}} = \mathbf{A}_1 \delta \mathbf{x} + \mathbf{B}_1 \delta \mathbf{u}$ where $\mathbf{A}_1 \in \mathbb{R}^{n \times n}$ and $\mathbf{B}_1 \in \mathbb{R}^{n \times n}$ are shown in (22), and

$$\begin{aligned} \delta \mathbf{x} &= [P_1 \quad P_2 \quad \dots \quad P_n]^T \in \mathbb{R}^n \\ \delta \mathbf{u} &= [a_{v1} \quad a_{v2} \quad \dots \quad a_{vn}]^T \in \mathbb{R}^n. \end{aligned} \quad (8)$$

² The state-space model representation is located at the end of the paper because of the size of the $\mathbf{A}$ and $\mathbf{B}$ matrices.

B. Architecture II

In Architecture II, (1), (2), (4), and (6) still hold. However, the common pressure of the refrigerant at the compressor inlet in Architecture II is P_1 , indicating that there is no pressure drop downstream of evaporator 1. Each of the $n-1$ DPR valves (see Fig. 2) is modeled as a variable resistor in a manner similar to that which was used for the n EEVs. The following $n-1$ algebraic equations represent the pressure drops across the DPR valves:

$$\begin{aligned} P_2 - P_1 &= R_{d2} \dot{m}'_2 - K_{d2} a_{d2} \\ P_3 - P_1 &= R_{d3} \dot{m}'_3 - K_{d3} a_{d3} \\ &\vdots \\ P_n - P_1 &= R_{dn} \dot{m}'_n - K_{dn} a_{dn} . \end{aligned}$$

Again, both R_{di} and K_{di} are positive constants arising from the linearization of the ME-VCC system, and a_{di} is the valve aperture of the i th DPR valve.

The condenser pressure can again be written as a linear combination of the n evaporator pressures, and we obtain the following proposition whose proof is similar to that of Proposition 1.

Proposition 2: The linear state-space model for Architecture II is not controllable. ■

Once again P_c is eliminated as a state from the system. By virtue of the $n-1$ DPR valves, there are $n-1$ additional control inputs in the system, so that

$$\delta \mathbf{u}_{II} = [\omega_k \quad a_{v1} \quad \cdots \quad a_{vn} \quad a_{d2} \quad \cdots \quad a_{dn}]^T \in \mathbb{R}^{2n}.$$

The state-space representation is given as $\mathbf{S}_{II} : \delta \dot{\mathbf{x}}_{II} = \mathbf{A}_{II} \delta \mathbf{x}_{II} + \mathbf{B}_{II} \delta \mathbf{u}_{II}$, $\mathbf{A}_{II} \in \mathbb{R}^{n \times n}$, $\mathbf{B}_{II} \in \mathbb{R}^{n \times 2n}$ and is shown in (23), where $\delta \mathbf{x}_{II}$ is the same as $\delta \mathbf{x}$ in (8). However, again $\text{rank}(\mathbf{B}_{II}) = n$ so only n of the $2n$ inputs independently affect the n states. Thus, for the purposes of analyzing and designing stabilizing controllers for this architecture, we will choose the same input vector $\delta \mathbf{u}$ as was chosen for Architecture I, shown in (8).³ We will show that these actuators are sufficient for designing decentralized controllers for this system, but our analysis can be applied to other choices of actuators as well.

III. DECENTRALIZED CONTROL ANALYSIS

In this section we will use the notion of connective stability [9] to provide conditions under which stabilizing decentralized controllers can be designed for arbitrarily large ME-VCC systems. We will start by decomposing the large ME-VCC system into a set of scalar, coupled subsystems and then analyze this representation to design local static feedback controllers that guarantee connective stability. To simplify the notation, the perturbation symbol δ is omitted in the subsequent analysis.

³ The additional n inputs may be useful when designing optimal controllers to regulate the mass flow and/or thermal dynamics in the ME-VCC system.

A. Subsystem Representation

The dynamics of the i th subsystem $\mathbf{S}_i$ (when considered in isolation) are defined as

$$\mathbf{S}_i : \dot{x}_i = A_i x_i + B_i u_i, \quad i=1,2,\dots,n \quad (9)$$

where $x_i \in \mathbb{R}$ is the refrigerant pressure (P_i) in the i th evaporator and $u_i \in \mathbb{R}$ is the valve aperture (a_{vi}) of the i th EEV. Next we define a control law, u_i , for the i th subsystem $\mathbf{S}_i$ as

$$u_i = -k_i x_i . \quad (10)$$

This control law is consistent with the design philosophy of ME-VCC systems in which it is desirable to control refrigerant flow to each evaporator independently of the flow to the other evaporators in the overall system. Substituting (10) into (9) yields

$$\mathbf{S}_i : \dot{x}_i = A_{i,CL} x_i = (A_i - B_i k_i) x_i, \quad i=1,\dots,n . \quad (11)$$

The interconnection between subsystems i and j is given by A_{ij} , $i \neq j$, the ij component of the overall system A-matrix $\mathbf{A}_I$ or $\mathbf{A}_{II}$. The overall system $\mathbf{S}$ is thereby decomposed as

$$\mathbf{S} : \dot{x}_i = A_{i,CL} x_i + \sum_{j=1}^n e_{ij} A_{ij} x_j \quad (12)$$

for $i, j=1,\dots,n$ where $e_{ij} \in [0,1]$ is an arbitrary function that captures variations and uncertainties in the coupling between subsystems i and j . These variations arise from model uncertainty and can be time-varying depending on how close the system is operating to the nominal operating condition about which it was linearized. Since the ME-VCC is actually a nonlinear system, this characterization of uncertainty in the coupling between subsystems is useful for ensuring that our linear system analysis is still relevant. We can now write $\mathbf{S}_I$ and $\mathbf{S}_{II}$ as decompositions of interconnected subsystems.

1) Architecture I

$\mathbf{S}_I$ is divided into n subsystems. The dynamics of the i th closed-loop subsystem are defined such that they are independent of parameters in any of the other j subsystems (where $j \neq i$) and are given by

$$\mathbf{S}_{1,i} : \dot{x}_{1,i} = A_{1,i,CL} x_{1,i} = \left(-\frac{1}{C_i} \left(\frac{1}{R_{vi}} + \frac{1}{R_{qi}} \right) - \frac{1}{C_c R_{vi}} - \frac{K_{vi} k_i}{C_i R_{vi}} \right) x_{1,i} .$$

The overall system $\mathbf{S}_I$ can be expressed as

$$\begin{aligned} \mathbf{S}_I : \dot{x}_{1,i} &= A_{1,i,CL} x_{1,i} + \sum_{j \neq i} \left(\frac{M}{C_i R_{qi}} \left(\frac{1}{R_{qj}} + \beta_{k2} \frac{C_j}{C_c} \right) - \frac{C_j}{C_c C_i R_{vi}} \right) x_{1,j} \\ &\quad + \left(\frac{M}{C_i R_{qi}^2} + \frac{M \beta_{k2}}{C_c R_{qi}} \right) x_{1,i} \end{aligned}$$

for all $i=1,\dots,n$.

2) Architecture II

$\mathbf{S}_{II}$ is also divided into n subsystems. The dynamics of the

i th closed-loop subsystem are defined as

$$\mathbf{S}_{\Pi,i} : \dot{x}_{\Pi,i} = \begin{cases} \left(-\frac{1}{C_i} \left(\frac{1}{R_{vi}} - \beta_{k1} \right) - \frac{1}{C_c} \left(\frac{1}{R_{vi}} - \beta_{k2} \right) - \frac{K_{vi} k_i}{C_i R_{vi}} \right) x_{\Pi,i}, & i=1 \\ \left(-\frac{1}{C_i} \left(\frac{1}{R_{vi}} + \frac{1}{R_{di}} \right) - \frac{1}{C_c R_{vi}} - \frac{K_{vi} k_i}{C_i R_{vi}} \right) x_{\Pi,i}, & i=2, \dots, n. \end{cases}$$

The overall system $\mathbf{S}_{\Pi}$ can be expressed as

$$\mathbf{S}_{\Pi} : \dot{x}_{\Pi,j} = A_{\Pi,j,CL} x_{\Pi,j} + \begin{cases} \sum_{j=2}^n -\frac{1}{C_i R_{dj}} x_{\Pi,j} + \sum_{j=2}^n \left(\left(\frac{C_j}{C_c C_i} \left(\beta_{k2} - \frac{1}{R_{vi}} \right) + \frac{1}{C_i R_{dj}} \right) x_{\Pi,j} \right), & i=1 \\ \left(\frac{1}{C_i R_{di}} - \frac{C_i}{C_c C_i R_{vi}} \right) x_{\Pi,1} - \sum_{j \neq 1}^n \left(\frac{C_j}{C_c C_i R_{vi}} x_{\Pi,j} \right), & i=2, \dots, n. \end{cases}$$

B. Connective Stability

Definition 1 ([9]): A system composed of subsystems of the form (12) is said to be connectively stable if $x_i(t) \rightarrow 0 \forall i \in \{1, \dots, n\}$ as $t \rightarrow \infty$ for any functions $e_{ij}(t) \in [0, 1], \forall i, j \in \{1, \dots, n\}, \forall t \in \mathbb{R}^+$.

This is a strong notion of stability because it captures state convergence to zero even with arbitrary (between 0 and 1) variations in the interconnection functions $e_{ij}(t)$. To show connective stability, we assume that each subsystem is stabilized with the local feedback controller given in (10), i.e. $A_{i,CL}$ in (11) is strictly negative. It follows that one can find an associated Lyapunov function $v_i(x_i)$ for the i th subsystem. Since each of the subsystems defined for $\mathbf{S}_I$ and $\mathbf{S}_{\Pi}$ are scalar, the Lyapunov function for the i th subsystem is defined as

$$v_i(x_i) = H_i x_i^2 \quad (13)$$

where H_i is obtained by solving the scalar Lyapunov equation $2A_{i,CL}H_i = -G_i$ for any $G_i > 0$. An appealing choice is $G_i = 1$, yielding $H_i = -(2A_{i,CL})^{-1} > 0$.

Here we want to show that when the individual subsystems are connected, stability is maintained. To do this, one attempts to find an appropriate Lyapunov function for the overall system. An appealing choice is

$$V(\mathbf{x}) = \sum_{i=1}^n \alpha_i v_i(x_i) \quad (14)$$

where the α_i 's, $i = 1, \dots, n$, are a set of positive constants.

The following test is provided in [9] to determine whether such a set of α_i 's exist. First a matrix $\hat{W}$ is defined whose elements are functions of the norms of the interconnection matrices and the individual Lyapunov functions defined in (13). Element (i, j) of this matrix can be obtained in a straightforward manner using the construction provided in Section 2.2 of [9], and is given by

$$\hat{w}_{ij} = \begin{cases} 1 + \frac{|A_{ii}|}{A_{i,CL}}, & i = j \\ \frac{|A_{ij}|}{A_{i,CL}}, & i \neq j. \end{cases} \quad (15)$$

This definition of $\hat{W}$ is valid for both $\mathbf{S}_I$ and $\mathbf{S}_{\Pi}$.

Lemma 1 ([9]): The system $\mathbf{S}$ is connectively stable if the eigenvalues of the matrix $\hat{W}$ lie in the open right half complex plane.

For $\hat{W}$ of the form in (15), we provide the following simpler test.

Lemma 2: The system $\mathbf{S}$ is connectively stable if

$$\sum_{j=1}^n |A_{ij}| < |A_{i,CL}| \quad \forall i \in \{1, \dots, n\}.$$

Proof: We will use Lemma 1 to prove Lemma 2. Specifically, we seek a condition on the scalar quantities A_{ij} , A_{ii} , and $A_{i,CL}$ that ensures that the eigenvalues of $\hat{W}$ lie in the open right half plane. To show Lemma 1 it is sufficient to show that all of the discs in the Gershgorin disc theorem [10] are contained within the open right half plane. This is equivalent to satisfying, for all $i \in \{1, \dots, n\}$,

$$\hat{w}_{ii} = 1 + \frac{|A_{ii}|}{A_{i,CL}} > 0$$

and

$$\sum_{j \neq i} |\hat{w}_{ij}| = \sum_{j \neq i} \frac{|A_{ij}|}{|A_{i,CL}|} < \hat{w}_{ii}.$$

After some algebra, we obtain

$$|A_{ii}| < |A_{i,CL}| \quad (16)$$

and

$$\sum_{j \neq i} |A_{ij}| < |A_{i,CL}| - |A_{ii}|. \quad (17)$$

By inspection we see that (17) implies (16), thereby proving the lemma. ■

Now we can use Lemma 2 to evaluate the connective stability of $\mathbf{S}_I$ and $\mathbf{S}_{\Pi}$.

1) Architecture I

Proposition 3: If every local feedback gain k_i satisfies the condition

$$k_i > \frac{1}{C_c R_{qi} K_{vi}} \sum_{j \neq i} \frac{|C_j R_{qi} R_{qj} - M R_{vi} (C_c + \beta_{k2} C_j R_{qj})|}{R_{qj}} + \frac{M R_{vi}}{C_c R_{qi}^2 K_{vi}} (C_c + \beta_{k2} C_i R_{qi}) - \frac{1}{C_c R_{qi} K_{vi}} (C_i R_{qi} + C_c R_{vi} + C_c R_{qi}) \quad (18)$$

for $i \in \{1, \dots, n\}$, then the closed-loop system $\mathbf{S}_I$ is connectively stable.

Proof: We will use Lemma 2 to prove Proposition 3. Specifically, we seek a condition on k_i that satisfies Lemma 2. This is equivalent to finding k_i satisfying

$$\sum_{j \neq i} \left| \frac{M}{C_i R_{qi}} \left(\frac{1}{R_{qj}} + \beta_{k_2} \frac{C_j}{C_c} \right) - \frac{C_j}{C_c C_i R_{ri}} \right| + \left| \frac{M}{C_i R_{qi}^2} + \frac{M \beta_{k_2}}{C_c R_{qi}} \right| < \left| -\frac{1}{C_i} \left(\frac{1}{R_{vi}} + \frac{1}{R_{qi}} + \frac{K_{vi} k_i}{R_{ri}} \right) - \frac{1}{C_c R_{vi}} \right|.$$

After some algebra, we obtain the lower bound in (18). ■

2) Architecture II

Proposition 4: If every local feedback gain k_i satisfies the conditions

$$k_1 > \frac{R_{v1}}{K_{v1}} \sum_{j \neq 1} \left| C_j C_c^{-1} (\beta_{k_2} - R_{v1}^{-1}) + R_{dj}^{-1} \right| + \frac{1}{C_c K_{v1}} \left(C_c R_{v1} \sum_{j \neq 1} R_{dj}^{-1} + C_1 (\beta_{k_2} R_{v1} - 1) \right) + \frac{(\beta_{k_1} R_{v1} - 1)}{K_{v1}} \quad (19)$$

and

$$k_i > \frac{1}{C_c K_{vi}} \left(\left(\sum_{j \neq i} C_j \right) - (C_c + C_i) \right), \quad i = 2, \dots, n \quad (20)$$

then the closed-loop system $\mathbf{S}_{II}$ is connectively stable.

Proof: The proof of Proposition 4 is similar to that of Proposition 3. The main exception is that the lower bound on k_1 is different than that on k_i for $i = 2, \dots, n$ because of the structure of $\mathbf{A}_{II}$, shown in (23), where the first row of the matrix is distinctly different from rows 2 through n . This is a consequence of the physical design of Architecture II in which there is no DPR valve located downstream of evaporator 1. ■

C. Discussion

An interesting feature of Architecture II is that the lower bound on k_1 in (19) is independent of the characteristics of EEVs 2 through n . This is consistent with physical intuition which suggests that the presence of DPR valves in the system decouples the dynamics of each EEV–evaporator flow path from one another. Perhaps more interestingly, we see that k_i for $i = 2, \dots, n$ (shown in (20)) is independent of *all* of the valve resistances in the system. This is very advantageous for ME-VCC systems in large buildings where the resistances can vary greatly depending on the lengths of pipe connecting the flow paths to one another, and hence highlights the robustness of Architecture II with respect to stability of the dynamics. Moreover, from (20) we see that if $C_c \geq \sum_j C_j$ ($j = 1, \dots, n$), then the lower bound on k_i ($i = 2, \dots, n$) is negative, indicating that *any* positive gain k_i will stabilize the system. This is a reasonable assumption for ME-VCC systems but would need to be verified for a given physical system as it depends on fluid characteristics such as bulk modulus [11].

On the other hand, for Architecture I, the bounds on the stability gains exhibit greater coupling with the other subsystems. While the lower bound on k_i shown in (18) is also independent of the characteristics of the other EEV resistances R_{vj} ($j \neq i$) in the system, k_i is dependent on *all* of the hydraulic resistances R_{qj} ($j = 1, \dots, n$) in the system. These hydraulic resistances are typically a function of the characteristics of the pipes connecting each of the evaporator outlet flow paths to one another.

Although it is possible to ensure connective stability under both system architectures, Architecture II is a more appealing choice for decentralized control of ME-VCC systems. This is due to 1) the decreased sensitivity of k_i ($i = 2, \dots, n$) (20) to parameters in other EEV–evaporator flow paths and 2) potentially small controller gains needed for stability as compared to Architecture I. While it is not surprising that the additional valves in Architecture II are useful for control of ME-VCC systems, our analysis shows how this benefit manifests itself in the controller gains.

IV. CONCLUSION

In this paper we considered the problem of stabilizing ME-VCC systems using decentralized controllers. By exploiting the time-scale separation characteristic of ME-VCC systems, a scalable analysis of the faster mass flow dynamics became possible. Using concepts from decentralized control theory, we designed local static feedback controllers for each evaporator so that the overall closed-loop system was stable and robust to uncertainties in the coupling between subsystems. The controllers were designed to provide connective stability, a strong result that guarantees stability even for time-varying uncertainties in the interconnections between subsystems. Therefore, within a sizable region around some nominal operating condition, we expect that the controllers can be used effectively on the original nonlinear ME-VCC system (despite being designed using a linear analysis). Finally, by analyzing two common but different ME-VCC system architectures, we gained valuable insight into the impact these architectures have on decentralized control design for these systems.

In future work, we will consider controller performance in addition to stability as well as practical concerns such as actuator saturation. Moreover, we will model the slower thermal dynamics in ME-VCC systems. Using a similar approach to that shown in this paper, we believe a scalable analysis of the thermal dynamics can be conducted with the ultimate goal of developing effective control strategies for these complex systems.

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$$\begin{bmatrix} \dot{P}_c \\ \dot{P}_1 \\ \dot{P}_2 \\ \vdots \\ \dot{P}_n \end{bmatrix} = \begin{bmatrix} \frac{1}{C_c} \left(\beta_{k2} (M \beta_{k1} + 1) - \sum_{i=1}^n \frac{1}{R_{qi}} \right) & \frac{1}{C_c} \left(\frac{1}{R_{v1}} - \frac{\beta_{k1} M}{R_{q1}} \right) & \frac{1}{C_c} \left(\frac{1}{R_{v2}} - \frac{\beta_{k1} M}{R_{q2}} \right) & \dots & \frac{1}{C_c} \left(\frac{1}{R_{vn}} - \frac{\beta_{k1} M}{R_{qn}} \right) \\ \frac{1}{C_1} \left(\frac{1}{R_{v1}} - \frac{1}{R_{q1}} M \beta_{k2} \right) & \frac{M}{C_1 R_{q1}^2} - \frac{1}{C_1} \left(\frac{1}{R_{v1}} + \frac{1}{R_{q1}} \right) & \frac{M}{C_1 R_{q1} R_{q2}} & \dots & \frac{M}{C_1 R_{q1} R_{qn}} \\ \frac{1}{C_2} \left(\frac{1}{R_{v2}} - \frac{1}{R_{q2}} M \beta_{k2} \right) & \frac{M}{C_2 R_{q2} R_{q1}} & \frac{M}{C_2 R_{q2}^2} - \frac{1}{C_2} \left(\frac{1}{R_{v2}} + \frac{1}{R_{q2}} \right) & \dots & \frac{M}{C_2 R_{q2} R_{qn}} \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ \frac{1}{C_n} \left(\frac{1}{R_{vn}} - \frac{1}{R_{qn}} M \beta_{k2} \right) & \frac{M}{C_n R_{qn} R_{q1}} & \frac{M}{C_n R_{qn} R_{q2}} & \dots & \frac{M}{C_n R_{qn}^2} - \frac{1}{C_n} \left(\frac{1}{R_{vn}} + \frac{1}{R_{qn}} \right) \end{bmatrix} \begin{bmatrix} P_c \\ P_1 \\ P_2 \\ \vdots \\ P_n \end{bmatrix} \quad (21)$$

$$+ \begin{bmatrix} (M \beta_{k1} + 1) \frac{\beta_{k3}}{C_c} & -\frac{K_{v1}}{C_c R_{v1}} & -\frac{K_{v2}}{C_c R_{v2}} & \dots & -\frac{K_{vn}}{C_c R_{vn}} \\ -M \frac{\beta_{k3}}{C_1 R_{q1}} & \frac{K_{v1}}{C_1 R_{v1}} & 0 & \dots & 0 \\ -M \frac{\beta_{k3}}{C_2 R_{q2}} & 0 & \frac{K_{v2}}{C_2 R_{v2}} & \ddots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ -M \frac{\beta_{k3}}{C_n R_{qn}} & 0 & 0 & 0 & \frac{K_{vn}}{C_n R_{vn}} \end{bmatrix} \begin{bmatrix} \omega_k \\ a_{v1} \\ a_{v2} \\ \vdots \\ a_{vn} \end{bmatrix}$$

$$A_1 = \begin{bmatrix} -\frac{1}{C_1} \left(\frac{1}{R_{v1}} + \frac{1}{R_{q1}} - \frac{M}{R_{q1}^2} \right) - \frac{1}{C_c} \left(\frac{1}{R_{v1}} - \frac{M \beta_{k2}}{R_{q1}} \right) & \frac{M}{C_1 R_{q1} R_{q2}} - \frac{C_2}{C_1 C_c} \left(\frac{1}{R_{v1}} - \frac{M \beta_{k2}}{R_{q1}} \right) & \dots & \frac{M}{C_1 R_{q1} R_{qn}} - \frac{C_n}{C_1 C_c} \left(\frac{1}{R_{v1}} - \frac{M \beta_{k2}}{R_{q1}} \right) \\ \frac{M}{C_2 R_{q2} R_{q1}} - \frac{C_1}{C_2 C_c} \left(\frac{1}{R_{v2}} - \frac{M \beta_{k2}}{R_{q2}} \right) & -\frac{1}{C_2} \left(\frac{1}{R_{v2}} + \frac{1}{R_{q2}} - \frac{M}{R_{q2}^2} \right) - \frac{1}{C_c} \left(\frac{1}{R_{v2}} - \frac{M \beta_{k2}}{R_{q2}} \right) & \ddots & \frac{M}{C_2 R_{q2} R_{qn}} - \frac{C_n}{C_2 C_c} \left(\frac{1}{R_{v2}} - \frac{M \beta_{k2}}{R_{q2}} \right) \\ \vdots & \ddots & \ddots & \vdots \\ \frac{M}{C_n R_{qn} R_{q1}} - \frac{C_1}{C_n C_c} \left(\frac{1}{R_{vn}} - \frac{M \beta_{k2}}{R_{qn}} \right) & \frac{M}{C_n R_{qn} R_{q2}} - \frac{C_2}{C_n C_c} \left(\frac{1}{R_{vn}} - \frac{M \beta_{k2}}{R_{qn}} \right) & \dots & -\frac{1}{C_n} \left(\frac{1}{R_{vn}} + \frac{1}{R_{qn}} - \frac{M}{R_{qn}^2} \right) - \frac{1}{C_c} \left(\frac{1}{R_{vn}} - \frac{M \beta_{k2}}{R_{qn}} \right) \end{bmatrix}, B_1 = \begin{bmatrix} \frac{K_{v1}}{C_1 R_{v1}} & 0 & \dots & 0 \\ 0 & \frac{K_{v2}}{C_2 R_{v2}} & \ddots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \dots & \frac{K_{vn}}{C_n R_{vn}} \end{bmatrix} \quad (22)$$

$$\begin{bmatrix} \dot{P}_1 \\ \dot{P}_2 \\ \dot{P}_3 \\ \vdots \\ \dot{P}_n \end{bmatrix} = \begin{bmatrix} \frac{1}{C_1} \left(-\frac{1}{R_{v1}} + \beta_{k1} - \sum_{i=2}^n \frac{1}{R_{di}} \right) + \frac{1}{C_c} \left(\beta_{k2} - \frac{1}{R_{v1}} \right) & \frac{C_2}{C_1 C_c} \left(\beta_{k2} - \frac{1}{R_{v1}} \right) + \frac{1}{C_1 R_{d2}} & \frac{C_3}{C_1 C_c} \left(\beta_{k2} - \frac{1}{R_{v1}} \right) + \frac{1}{C_1 R_{d3}} & \dots & \frac{C_n}{C_1 C_c} \left(\beta_{k2} - \frac{1}{R_{v1}} \right) + \frac{1}{C_1 R_{dn}} \\ \frac{1}{C_2 R_{d2}} - \frac{C_1}{C_2 C_c R_{v2}} & -\frac{1}{C_2} \left(\frac{1}{R_{v2}} + \frac{1}{R_{d2}} \right) - \frac{1}{C_c R_{v2}} & -\frac{C_3}{C_2 C_c R_{v2}} & \dots & -\frac{C_n}{C_2 C_c R_{v2}} \\ \frac{1}{C_3 R_{d3}} - \frac{C_1}{C_3 C_c R_{v3}} & -\frac{C_2}{C_3 C_c R_{v3}} & -\frac{1}{C_3} \left(\frac{1}{R_{v3}} + \frac{1}{R_{d3}} \right) - \frac{1}{C_c R_{v3}} & \ddots & -\frac{C_n}{C_3 C_c R_{v3}} \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ \frac{1}{C_n R_{dn}} - \frac{C_1}{C_n C_c R_{vn}} & -\frac{C_2}{C_n C_c R_{vn}} & -\frac{C_3}{C_n C_c R_{vn}} & \dots & -\frac{1}{C_n} \left(\frac{1}{R_{vn}} + \frac{1}{R_{dn}} \right) - \frac{1}{C_c R_{vn}} \end{bmatrix} \begin{bmatrix} P_1 \\ P_2 \\ P_3 \\ \vdots \\ P_n \end{bmatrix}$$

$$+ \begin{bmatrix} \frac{\beta_{k3}}{C_1} & \frac{K_{v1}}{C_1 R_{v1}} & 0 & \dots & 0 & \frac{K_{d2}}{C_1 R_{d2}} & \frac{K_{d3}}{C_1 R_{d3}} & \dots & \frac{K_{dn}}{C_1 R_{dn}} \\ 0 & 0 & \frac{K_{v2}}{C_2 R_{v2}} & \dots & 0 & \frac{K_{d2}}{C_2 R_{d2}} & 0 & \dots & \dots \\ 0 & 0 & 0 & \ddots & 0 & 0 & \frac{K_{d3}}{C_3 R_{d3}} & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \frac{K_{vn}}{C_n R_{vn}} & 0 & 0 & \dots & -\frac{K_{dn}}{C_n R_{dn}} \end{bmatrix} \begin{bmatrix} \omega_k \\ a_{v1} \\ a_{v2} \\ \vdots \\ a_{vn} \\ a_{d2} \\ a_{d3} \\ \vdots \\ a_{dn} \end{bmatrix} \quad (23)$$